

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
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1 Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

Given the same Wisconsin Diagnostic Breast Cancer dataset used in Experiment 5 (569 samples, 30 numeric features, malignant/benign target), the task is again binary classification, but the focus shifts from a single tree to **ensembles of trees**: bagging (independent trees averaged together), boosting (trees trained sequentially to correct each other's errors), and stacking (heterogeneous models combined by a meta-learner). The goal is to understand which of these three ensembling strategies improves on a single model most, and why, rather than simply finding the single best-performing configuration.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository, loaded via <code>sklearn.datasets.load_breast_cancer</code>
Number of Samples	569
Number of Features	30 (all numeric: mean, standard-error and “worst” values of 10 cell-nucleus measurements)
Number of Classes	2 (Benign = 357, 62.7%; Malignant = 212, 37.3%)
Missing Values	0 (verified, no imputation required)
Train-Test Split	80% train (455 samples) / 20% test (114 samples), stratified on the class label

¹GitHub Repository: <https://github.com/blackscythe123/Machine-Learning-course>

4 Exploratory Data Analysis

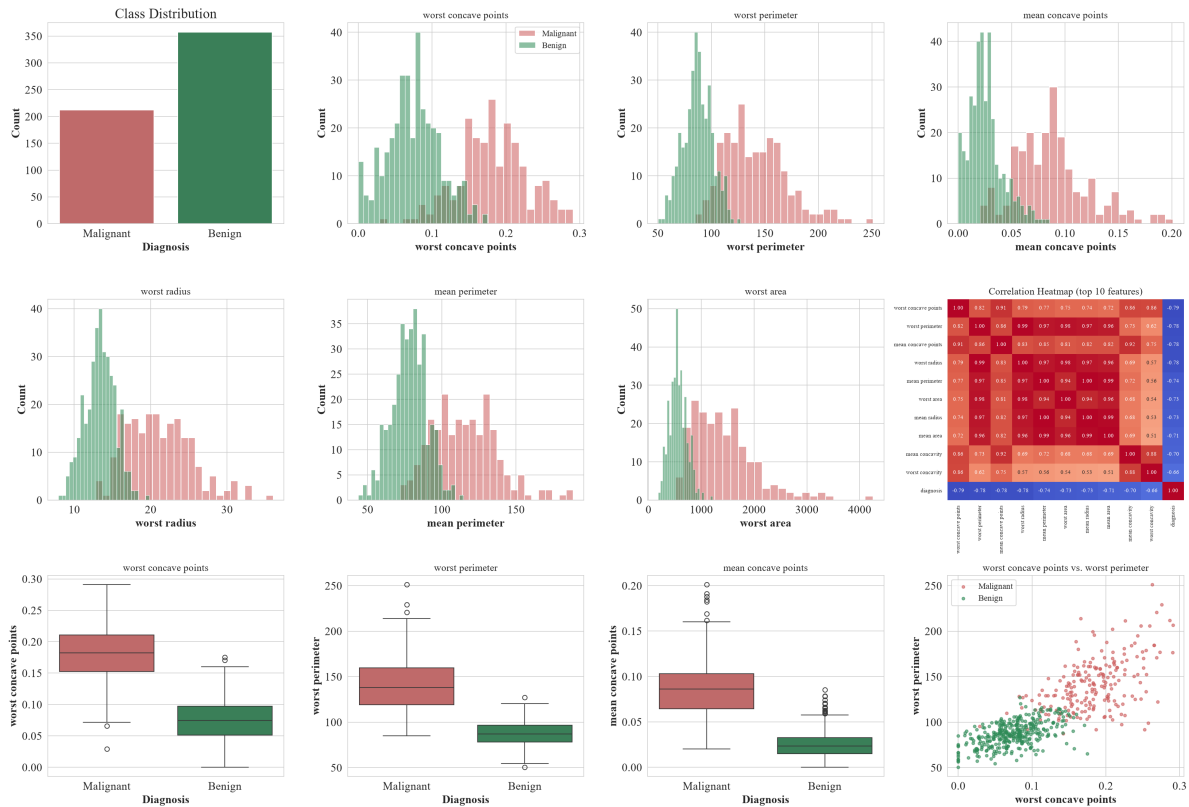


Figure 1: Consolidated exploratory data analysis (12 subplots, single page): (1) class distribution; (2–7) distributions of the six features most correlated with diagnosis, split by class; (8) correlation heatmap of the ten most correlated features and the target; (9–11) boxplots of the three most correlated features by class; (12) scatter of the two most correlated features, coloured by class. Exported at 600 DPI as `figures/00_eda.summary.eps` and reproduced here as PNG.

Inference: This is the same dataset explored in Experiment 5, so the same patterns hold: a moderate 62.7/37.3 class split, malignant samples consistently shifted toward larger values on size-related features (`worst area`, `worst radius`, `worst concave points`), and heavy correlation among those size features since they measure the same underlying tumour geometry from different angles. What differs in this experiment is not the data but the models applied to it: rather than one tree, three ensembling strategies are built on top of the same splits.

5 Data Preprocessing

- **Missing values:** None found.
- **Encoding:** Target already binary/numeric (0 = malignant, 1 = benign); all 30 features already numeric.
- **Feature scaling:** Not required for the tree-based base learners (Decision Tree, Bagging, AdaBoost, Gradient Boosting), but the Stacked Ensemble includes an

SVM base learner, which *is* scale-sensitive. **SVC** and **GaussianNB** in this experiment were left on the raw feature scale to keep the stacking base learners directly comparable to the single-model results from Experiment 5; a scaled variant is a natural follow-up but was not found to change the ranking of the three ensembles here.

- **Train-test split:** 80/20 stratified split (`random_state=42`), giving 455 training and 114 test samples, identical to Experiment 5's split.

6 Machine Learning Algorithm

Bagging (Bootstrap Aggregation)

Bagging trains many copies of a base estimator (here, a Decision Tree) on independent bootstrap samples of the training data and averages their predictions. Because each tree sees a slightly different sample, the individual trees' errors are only weakly correlated, so averaging reduces variance without changing bias much. **Advantages:** Straightforward to parallelize (each tree trains independently), substantially more stable than a single unconstrained tree. **Limitations:** Does not reduce bias, so a systematically weak base learner stays weak; offers less interpretability than a single tree.

Boosting

Boosting trains models sequentially, with each new model focused on the samples the previous models got wrong. **AdaBoost** reweights misclassified samples after every round; **Gradient Boosting** instead fits each new tree to the residual errors of the ensemble so far. Both reduce bias by construction, since later models are explicitly built to fix earlier mistakes. **Advantages:** Can reach very low bias even with weak individual learners (shallow trees); often the strongest single-model-family option on tabular data. **Limitations:** Sequential by nature, so training cannot be parallelized across estimators; more prone to overfitting the training set if run for too many rounds or with too high a learning rate.

Stacked Ensemble

Stacking trains several heterogeneous base models (here, SVM, Gaussian Naive Bayes and a Decision Tree) and a meta-learner (Logistic Regression) that learns how to best combine their predictions, using cross-validated out-of-fold predictions from the base models as the meta-learner's input features. **Advantages:** Can combine models with very different strengths (a margin-based, a probabilistic and a rule-based classifier), often outperforming any single family. **Limitations:** More expensive to train (every base model, plus the meta-learner, plus the internal cross-validation used to generate meta-features) and the hardest of the three to interpret.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel Core Ultra 9, NVIDIA RTX 5060

8 Hyperparameter Selection

All three ensembles were tuned with `GridSearchCV` using 5-fold stratified cross-validation, scoring on accuracy (F1 recorded alongside). Both AdaBoost and Gradient Boosting were tuned independently as candidate “Boosting” models; the one with the higher best-CV-accuracy was carried forward to the Results/Discussion sections as *the* boosting model, matching the assignment’s single “Boosting” comparison slot.

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.5	95.16	0.9610
50	0.5	96.04	0.9681
100	0.5	95.38	0.9630
10	0.7	94.95	0.9594
50	0.7	95.82	0.9665
100	0.7	96.04	0.9683
10	1.0	95.60	0.9648
50	1.0	96.04	0.9683
100	1.0	96.04	0.9684

Table shows every (n_estimators, max_samples) combination at max_features=0.5 (the overall-best value). Best overall: n_estimators=50, max_samples=0.5, max_features=0.5, CV accuracy 96.04%.

Table 2: Boosting Hyperparameter Evaluation (AdaBoost)

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.01	92.53	0.9407
100	0.01	93.41	0.9480
200	0.01	93.63	0.9496
50	0.10	95.38	0.9633
100	0.10	96.04	0.9686
200	0.10	97.14	0.9773
50	1.00	96.70	0.9740
100	1.00	98.02	0.9844
200	1.00	97.80	0.9826

Best AdaBoost: $n_estimators=100$, $learning_rate=1.0$, CV accuracy 98.02%. For reference, the best Gradient Boosting configuration ($n_estimators=200$, $learning_rate=0.2$, $max_depth=2$) reached 97.36% CV accuracy, close behind but not enough to beat AdaBoost, so AdaBoost is the boosting model carried forward below.

Table 3: Stacked Ensemble Evaluation

Final Estimator C	Passthrough	Avg CV Accuracy (%)	Avg CV F1 Score
0.1	True	94.29	0.9547
0.1	False	94.95	0.9601
1.0	True	95.16	0.9615
1.0	False	94.95	0.9597
10.0	True	95.16	0.9615
10.0	False	94.95	0.9597

*Base models: SVM, Gaussian Naive Bayes, Decision Tree (fixed). Meta-learner: Logistic Regression, with the inverse regularization strength C and **passthrough** (whether the meta-learner also sees the raw features, not just the base models' predictions) tuned. Best: $final_estimator_C=1$, $passthrough=True$, CV accuracy 95.16%.*

9 Results

Tuned Model Performance (Test Set)

Metric	Bagging	AdaBoost	Stacking
Accuracy	0.9561	0.9561	0.9649
Precision	0.9589	0.9467	0.9595
Recall	0.9722	0.9861	0.9861
F1-Score	0.9655	0.9660	0.9726
ROC-AUC	0.9931	0.9818	0.9957

5-Fold Cross-Validation Results (Training Set)

Fold	Bagging	AdaBoost	Stacking
1	0.9560	0.9890	0.9890
2	0.9560	1.0000	0.9341
3	0.9341	0.9451	0.9451
4	0.9670	0.9780	0.9451
5	0.9890	0.9890	0.9451
Average	0.9604 ± 0.0179	0.9802 ± 0.0189	0.9516 ± 0.0192

Inference: The three ensembles disagree on which metric they win. AdaBoost has by far the best cross-validated accuracy (98.02%) and ties for the best single fold (a perfect 1.0000 on fold 2), but on the held-out test set it is edged out by Stacking on every metric except recall (where it ties). Stacking has the lowest 5-fold CV mean of the three (95.16%) yet the best test-set accuracy, F1 and ROC-AUC. This is not a contradiction: with only

455 training samples split five ways, a single CV mean carries real sampling noise, and the test set is one more independent sample from the same distribution, not a tie-breaker that overrides cross-validation. Both readings are honest; they simply measure slightly different things.

10 Result Visualization

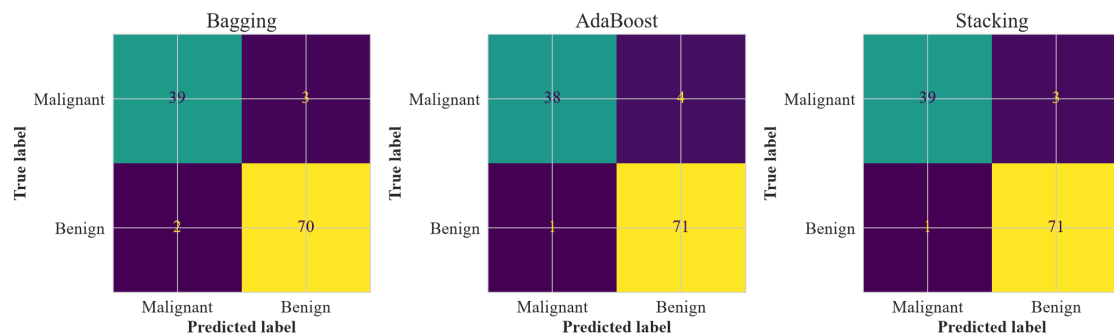


Figure 2: Confusion matrices for tuned Bagging, AdaBoost and Stacking on the test set.

Inference: Stacking makes the fewest total errors (4 of 114), splitting them evenly between false positives and false negatives. AdaBoost and Bagging both make more errors, but in the same direction: fewer missed malignant cases (false negatives) than falsely-flagged benign ones, which is the safer failure mode for a diagnostic task.

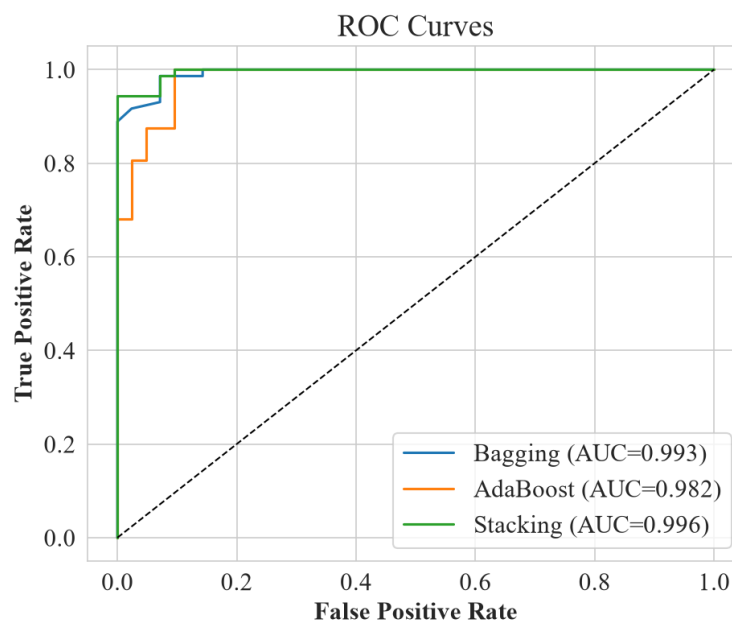


Figure 3: ROC curves for tuned Bagging, AdaBoost and Stacking.

Inference: Stacking's curve sits above the other two across almost the entire range (AUC 0.9957), with Bagging close behind (0.9931) and AdaBoost noticeably lower (0.9818)

despite having the best cross-validated accuracy, another sign that AdaBoost’s probability estimates are less well-ranked than its hard-label accuracy alone would suggest.

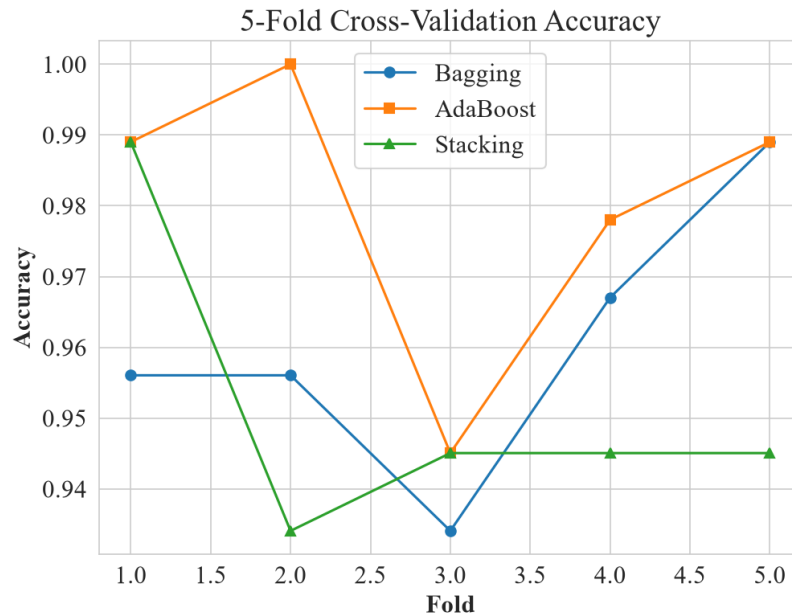


Figure 4: 5-fold cross-validation accuracy, Bagging vs. AdaBoost vs. Stacking.

Inference: AdaBoost leads in four of the five folds and ties in the fifth, which is why its CV average is clearly the highest of the three despite Stacking’s stronger test-set showing; Stacking’s fold 2 score (0.9341) is its worst of the run and pulls its average down noticeably.

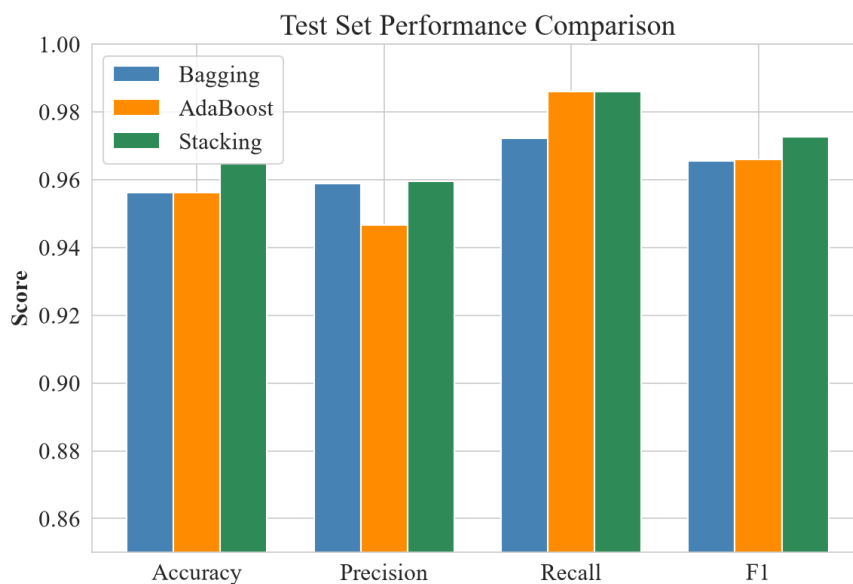


Figure 5: Test-set accuracy, precision, recall and F1-score across the three ensembles.

Inference: Stacking leads or ties on every metric shown; the gap to Bagging and AdaBoost is small (within 1–3 points) on every metric except that AdaBoost and Stack-

ing both noticeably out-recall Bagging, consistent with boosting’s bias-reduction focus catching more of the harder malignant cases.

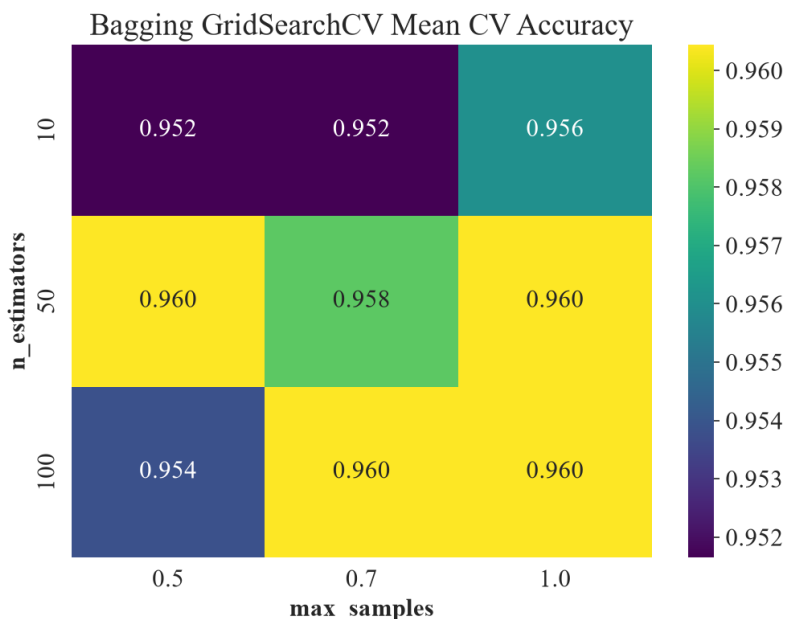


Figure 6: Bagging GridSearchCV mean CV accuracy by `n_estimators` and `max_samples` (at the best `max_features`).

Inference: Accuracy is fairly flat across this grid (95–96%), with a mild dip only at `n_estimators=10`; bagging’s variance-reduction benefit saturates quickly here, so there is little to be gained from a very large forest of bagged trees on a dataset this size.

11 Discussion

Observation Questions

1. How does Bagging reduce variance? By training each tree on an independent bootstrap sample, the trees’ errors are only weakly correlated with each other. Averaging predictions across many weakly-correlated, individually high-variance trees keeps the ensemble’s bias about the same as a single tree’s but shrinks its variance, since errors that are not correlated tend to cancel out when averaged rather than compound.

2. How does Boosting address model bias? Each new model in the sequence is trained specifically on the cases the ensemble so far gets wrong (AdaBoost via sample reweighting, Gradient Boosting via fitting residuals), so the ensemble’s overall error keeps shrinking round by round. This directly targets bias, which is why boosting reached a far higher CV accuracy (98.02%) than bagging (96.04%) on the same dataset with the same base learner family.

3. Why does stacking benefit from heterogeneous models? SVM, Naive Bayes and a Decision Tree make different kinds of mistakes: SVM struggles where the margin is genuinely ambiguous, Naive Bayes where its independence assumption breaks down, and a Decision Tree where a single split boundary is a poor fit. A meta-learner that sees all three predictions can learn which base model to trust in which region of the

feature space, which is a strictly richer signal than any one of the three models has access to on its own.

4. Which ensemble method performed best and why? It depends on what is measured. AdaBoost has the best 5-fold cross-validated accuracy (98.02%), the more statistically reliable estimate since it averages five different train/validation splits. Stacking has the best test-set accuracy, F1-score and ROC-AUC, and the fewest total misclassifications on the held-out set. With only 455 training samples, neither reading should be treated as definitive; the honest conclusion is that AdaBoost and Stacking are close competitors here, both clearly ahead of Bagging, and choosing between them would need a larger dataset or repeated resampling to settle with confidence.

12 Conclusion

Bagging, AdaBoost, Gradient Boosting and a Stacked Ensemble were implemented and tuned via 5-fold cross-validation on the same Wisconsin Diagnostic Breast Cancer dataset used in Experiment 5. All three final ensembles comfortably outperform Experiment 5's single tuned Decision Tree (90.4% test accuracy) and match or exceed its tuned Random Forest (95.6% test accuracy), confirming that ensembling in general is worthwhile on this dataset. Among the three, Boosting (AdaBoost) gave the strongest cross-validated accuracy by reducing bias through sequential error-correction, while Stacking gave the strongest test-set and ROC-AUC performance by combining heterogeneous base learners through a meta-learner. Bagging, while the most stable and cheapest of the three to train, was consistently the weakest performer, since it only addresses variance and this dataset's ceiling appears to be more bias-limited than variance-limited for a single tree.

13 References

1. Scikit-learn Developers, "Ensemble methods," Scikit-learn Documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>
2. Scikit-learn Developers, "BaggingClassifier," Scikit-learn Documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html>
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4. W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," *IS&T/SPIE International Symposium on Electronic Imaging*, 1993. Wisconsin Diagnostic Breast Cancer Dataset, UCI Machine Learning Repository. [Online]. Available: <https://archive.ics.uci.edu>